

1. Cell Class

- **Attributes:**

- Content: Object: Represents what is inside the cell (could be a reward, punishment, enemy, or obstacle).
- isWalkable: boolean: Determines if Doug or the enemies can move through this cell.

- **Methods:**

- setContent(content: Object): void: Assigns content to the cell (e.g., placing a reward, enemy, or obstacle).
- clearContent(): void: Clears the cell's content (e.g., when Doug collects a reward).
- isBlocked(): boolean: Returns true if the cell is blocked by a wall, barrier, or obstacle and cannot be passed through.

2. Doug Class

- **Attributes:**

- positionX: int: Doug's current X-coordinate on the grid.
- positionY: int: Doug's current Y-coordinate on the grid.
- score: int: The current score Doug has accumulated from rewards.
- collectedFood: List<Reward>: A list of all the rewards Doug has collected so far.
- health: int: Doug's remaining health, which is reduced by punishments or enemy interactions.

- **Methods:**

- move(direction: String): void: Moves Doug one step in the specified direction (up, down, left, right).
- collectReward(reward: Reward): void: Adds a reward to Doug's score and list of collected food.
- penalize(punishment: Punishment): void: Applies a penalty (reduces Doug's health or score).
- isGameOver(): boolean: Returns true if Doug's health reaches zero or an enemy catches him, ending the game.

3. MovingEnemy Class

- **Attributes:**

- positionX: int: Enemy's current X-coordinate on the grid.
- positionY: int: Enemy's current Y-coordinate on the grid.

- Target: Doug: Reference to Doug, whom the enemy is trying to chase.
- **Methods:**
 - move(): void: Moves the enemy one step in its current direction.
 - chase(Doug doug): void: The enemy calculates the best path towards Doug and moves one step in his direction.

4. Punishment Class

- **Attributes:**
 - penaltyPoints: int: The number of points or health that will be subtracted when Doug interacts with the punishment.
 - Type: String: Describes the type of punishment (e.g., "Chocolate" or "Onion").
- **Methods:**
 - applyPenalty(Doug doug): void: Applies the penalty to Doug, reducing his score or health when he lands on a punishment cell.

5. Reward Class

- **Attributes:**
 - Points: int: The points Doug gains by collecting this reward.
 - Type: String: Describes the type of reward (e.g., "Hot Dog" or "Sushi").
 - Image: String: A reference to the image representing the reward in the game.
- **Methods:**
 - collect(): int: When Doug collects the reward, it returns the number of points to add to his score.

6. BonusReward Class (Subclass of Reward)

- **Attributes:**
 - bonusPoints: int: Additional points Doug can gain by collecting a bonus reward.
- **Methods:**
 - collectBonus(): int: Similar to collect(), but applies a higher bonus score when collected.

7. Enemy Class

- **Attributes:**
 - positionX: int: Enemy's current X-coordinate on the grid.
 - positionY: int: Enemy's current Y-coordinate on the grid.

- Type: String: Describes the type of enemy (e.g., "Cat" or "Rat").
- Image: String: Reference to the enemy's image in the game.
- **Methods:**
 - move(): void: Moves the enemy one step on the grid.
 - interact(Doug doug): void: Defines how the enemy interacts with Doug. For example, if the enemy catches Doug, the game is over.

8. Cat Class (Subclass of Enemy)

- **Methods:**
 - move(): void: Moves the cat one step on the grid.
 - stealFood(Doug doug): void: If the cat reaches Doug, it "steals" all his food, causing Doug to lose the game.

9. Rat Class (Subclass of Enemy)

- **Methods:**
 - move(): void: Moves the rat one step on the grid.
 - stealFood(Doug doug): void: Similar to the cat, if the rat reaches Doug, it steals all his food, leading to an immediate game over.

10. Item Class

- **Attributes:**
 - Points: int: The points associated with the item (could be a reward or punishment).
 - Image: String: The image representing the item on the game board.
- **Methods:**
 - collect(): int: When Doug collects the item, this method returns the points to be added to his score.

11. Obstacle Class

- **Attributes:**
 - Type: String: Describes the type of obstacle (e.g., "Wall", "Tree").
- **Methods:**
 - blockMovement(): boolean: Returns true if the obstacle blocks Doug or an enemy from moving through the cell it occupies.

12. Map Class

- **Attributes:**
 - Cells: 2D array<Cell>: Represents the game map as a 2D grid of cells.

- mapName: String: The name of the current map or level (e.g., "Grasslands").
- startPoint: Cell: The starting point of the game where Doug begins.
- endPoint: Cell: The endpoint of the game that Doug must reach to win.
- **Methods:**
 - loadMap(int level): void: Loads the map for a specific level.
 - renderMap(): void: Renders the game map on the screen, showing all the cells and their content.
 - updateCell(int x, int y, Object content): void: Updates the content of a specific cell (e.g., adding or removing a reward, enemy, etc.).

13. GameMechanics Class

- **Attributes:**
 - Score: int: The current score in the game.
 - TimeElapsed: int: The time that has passed since the start of the game.
 - gameState: String: Represents the state of the game (e.g., "Running", "GameOver", "Victory").
- **Methods:**
 - startGame(): void: Initializes and starts the game.
 - endGame(): void: Ends the game and displays the game over screen.
 - checkVictory(): boolean: Checks if Doug has won the game by collecting all rewards and reaching the exit.
 - checkLoss(): boolean: Checks if Doug has lost the game (e.g., caught by an enemy, health reaches zero).